

Is Prompt Rewriting Always Beneficial? A Benchmark for Quantifying Prompt Informativeness and Stability in Text-to-Image Models

Anonymous submission

Abstract

Recent text-to-image (T2I) generative models have enabled users to produce strikingly realistic images from natural language prompts. However, prompt informativeness varies significantly across user proficiency levels; e.g., ambiguous or under-specified prompts often lead to unstable outputs that deviate from user intent. Although existing methods commonly assume that prompt rewriting improves generation, this assumption has received limited systematic investigation. Not every prompt benefits from optimization: *unnecessary rewriting may inject noise and hallucinate unsupported details, thus deviating from the user’s original intent*. Thus, we introduce **Authentic Prompt Benchmark (AP Bench)**, a large-scale benchmark of 17,580 authentic intent–prompt–image triplets sourced from real-world web repositories, spanning from novice (e.g., short and informal words) to expert (e.g., highly detailed, professionally composed specifications) users. Unlike existing metrics that focus on prompt–image alignment, we position AP Bench as the first dedicated benchmark for investigating prompt-to-prompt transmission and evaluating user-oriented prompt stability in T2I generation. Building on the insights from our AP Bench, we further propose **NoxEye**, a novel end-to-end prompt optimization framework for enhancing T2I generation. Across AP Bench and other established benchmarks, NoxEye delivers improvements of up to 16.09% in mutual information, 18.99% in prompt entropy, and 24.49% in prompt energy. Importantly, we demonstrate that NoxEye can not only genuinely improve authentic prompts written by real users, but also know how to selectively refine them while faithfully preserving the users’ original intent, serving as a plug-and-play framework that consistently boosts the performance of existing state-of-the-art T2I generative models, as also verified by human evaluations.

Code and Datasets — <https://authpromptbench.github.io/>

Introduction

Text-to-image (T2I) models (Rombach et al. 2022; Ho, Jain, and Abbeel 2020; Ramesh et al. 2021, 2022; Chen, Ge et al. 2024; Jiang et al. 2024; Lee et al. 2022; Han et al. 2025; Xie, Yang, and Shou 2025; Chen et al. 2025) convert natural-language prompts into visual outputs, but their reliability

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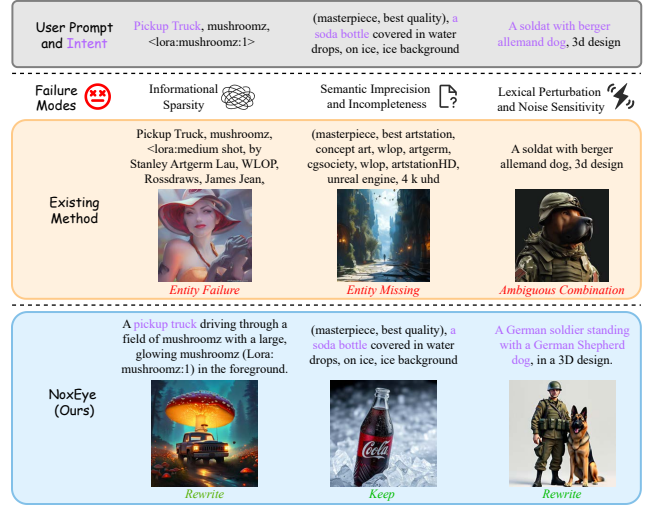


Figure 1: A Taxonomy of Prompt-induced **Instability**. We highlight three failure modes arising from different user prompt patterns with a baseline prompt optimization method (Hao et al. 2023), and show that our method yields **more stable and intent-aligned** generations. This suggests that an effective prompt optimization method should not only improve **1. prompt informativeness** to facilitate generation, but also preserve **2. generation stability** and faithfully retain the user’s intent.

depends on how effectively a prompt encodes the user’s intended content. Our rationale is motivated by the observation that most prior studies on prompt refinement (Hao et al. 2023; Cao et al. 2023; Rosenman, Lal, and Howard 2023; Wang et al. 2025) primarily focus on improving text–image alignment or aesthetic quality, while largely overlooking a fundamental question: **whether prompts written by real users can faithfully convey their underlying intent, and whether rewriting is actually necessary**. In practice, user prompting proficiency is highly heterogeneous, and such variability in prompt informativeness can substantially affect the stability of text-to-image generation. For example, expert users often craft long and detailed prompts with explicit attributes, whereas novice users tend to provide shorter, underspecified, and ambiguous descriptions.

Although recent approaches leverage LLMs to enhance or rewrite user prompts for improved text–image alignment



Figure 2: We observe that state-of-the-art generative models, such as Stable Diffusion 2.1 (SD2.1) (Rombach et al. 2022) and CoMat 2.1 (CoMat) (Jiang et al. 2024), remain prone to generation instability under authentic user inputs. Our selective prompt-refinement method, NoxEye, mitigates this limitation by systematically enhancing prompt–image alignment with user intent and outperforms the state-of-the-art prompt-refining method, e.g., Promptist (Hao et al. 2023) and PAG (Yun et al. 2025). Instead of **stylistic injection** caused by mode collapse (Yun et al. 2025), our approach focuses on **semantic clarification** and **avoiding hallucinations**.

(Hao et al. 2023; Yun et al. 2025; Mo et al. 2024), they typically assume that rewriting is universally beneficial. However, unconditional refinement can alter the original intent and introduce prompt-induced instability. As illustrated in Figure 1, we identify three common **failure modes** in real-world prompts: (i) informationally sparse prompts may trigger unsupported expansions; (ii) semantically incomplete prompts may result in partial intent preservation; and (iii) lexical noise may cause semantic drift or incorrect interpretations (Du et al. 2023; Chefer et al. 2023). These observations indicate that prompt optimization is not merely an alignment problem, but a selective intent-preservation problem. Therefore, effective T2I prompt optimization requires not only better rewriting strategies but also a mechanism to determine when rewriting should be applied.

Therefore, we cast prompt stabilization as a selective refinement problem: the goal is to generate intent-preserving rewrites while avoiding unnecessary modifications to prompts that already express the target intent. To this end, we propose **NoxEye**, a plug-and-play prompt optimization framework that learns both how to refine a prompt and whether the refinement should be applied for a target T2I model. NoxEye is trained in two stages: (1) supervised fine-tuning to generate intent-preserving candidate refinements from authentic prompts; and (2) rewrite-utility learning to predict whether a refinement improves intent preservation based on model-specific generation feedback. This design enables NoxEye to selectively enhance ambiguous prompts while retaining already effective ones.

Despite increasing attention to prompt optimization, we identify a fundamental gap in the community: **the lack of a systematic benchmark for evaluating user-side prompt informativeness and stability under realistic conditions**. To facilitate research in this area, we introduce the Authentic Prompt Benchmark, the first dedicated benchmark for sys-

tematically evaluating user-side prompt optimization methods in text-to-image generation under realistic prompting scenarios. Built from 17,580 authentic intent–prompt–image triplets collected from real-world prompt-sharing platforms, AP Bench moves beyond manually designed or synthetic prompts and captures the diversity, ambiguity, and uneven quality of practical user inputs. Among them, 2,048 representative samples are carefully annotated for benchmarking and further divided into *novice* and *expert* subsets according to prompt design proficiency (Stanford University Information Technology 2026; Meta 2026; LetsEnhance 2026). To evaluate T2I stability, we design a multi-level evaluation framework covering prompt-side informativeness and image-side intent preservation, including entropy and energy measures, CLIP-based intent recovery, MLLM-based semantic evaluation, and human assessment.

Extensive experiments demonstrate that NoxEye effectively mitigates the adverse effects of ambiguous prompts, substantially improving both fidelity and stability of generated images. When compared to the state-of-the-art prompt optimization strategies, our framework yields 16.09%, 18.99% and 24.49% gains in mutual information, prompt entropy, and prompt energy metrics. Meanwhile, NoxEye can advance generative ability on existing T2I models. Our framework excels both in GenEval (+2.16%) and T2I-CompBench (+5.11%).

Our contributions are threefold. First, we formulate prompt stability as semantic intent preservation through the information channel, distinguishing it from pixel-level similarity or generation variance alone. Second, we introduce AP Bench, the first intent-conditioned benchmark built from authentic prompt–image pairs to evaluate prompt optimization beyond image quality, revealing failures of unconditional rewriting in preserving user intent. Third, we propose NoxEye, a target-model-aware selective refinement framework that learns when to rewrite prompts and when to retain.

Related Work

Text-to-Image Generation and Benchmarks

Text-to-image generation has progressed from early GANs (Goodfellow et al. 2014) and VAEs (Kingma and Welling 2013) to diffusion-based models, with Stable Diffusion (Rombach et al. 2022) and CoMat (Jiang et al. 2024) exemplifying the current paradigm. These models typically employ frozen text encoders (e.g., CLIP (Radford et al. 2021)) to map prompts to embeddings that guide iterative denoising.

Benchmarking efforts have evolved alongside model capabilities. HEIM (Lee, Yasunaga et al. 2023) evaluates twelve dimensions, including alignment, quality, reasoning, and fairness. T2I-CompBench (Huang et al. 2025) focuses on compositional generation with novel metrics and reward-driven fine-tuning (GORS), while GenEval (Ghosh, Hajishirzi, and Schmidt 2023) introduces object-centric evaluation for fine-grained analysis. Existing benchmarks assess quality, object correctness, or compositional alignment with curated prompts, but lack predefined intent for authentic prompt-image pairs. AP Bench introduces an intent-first protocol to directly evaluate intent preservation in real-world prompts.

Prompt Optimization for Text-to-Image Generation

Some researchers have noticed that prompt design plays an essential role in making the model better understand our intentions and producing higher-quality results (Hao et al. 2023). Simultaneously, prompt optimization leverages LLMs (Yun et al. 2025) to improve generated image quality. Promptist (Hao et al. 2023) fine-tunes GPT-2 (Radford et al. 2019) to reformulate user prompts via supervised fine-tuning (SFT) and direct preference optimization (DPO) using CLIP similarity and aesthetics scores. PromptCoT (Yao et al. 2024) and RePrompt (Wu et al. 2026) incorporates the Chain-of-Thought (CoT) mechanism to learn high-quality prompt expressions. Self-Rewarding LVLs (Yang et al. 2025) extend this two-stage paradigm with a self-reward mechanism, while PAG (Yun et al. 2025) uses GFlowNets to generate diverse adaptive prompts. OPT2I (Mañas et al. 2024), TIR (Khan et al. 2025), and RATTPO (Kim et al. 2025) iteratively refine prompts by repeatedly generating images and updating the prompts based on the resulting feedback.

These approaches enhance prompt quality but largely focus on aesthetic and relevance objectives, often neglecting whether generated images faithfully reflect the user’s underlying intent, the illusion problem of large language models (Kalai et al. 2025) and time cost (Venkatesh et al. 2025).

In addition, prior studies have explored prompt optimization for image generation through prompt embeddings and dynamically controlled prompts. For example, LLM4GEN (Liu et al. 2025) enhances CLIP embeddings by leveraging representations from large language models, thereby improving the generative performance of Stable Diffusion (Stability AI 2023) models, while PAE (Mo et al. 2024) employs dynamically controlled prompts to guide the denoising process for more refined image synthesis. Although these approaches demonstrate effectiveness in specific settings, their generalizability is limited, and they cannot be readily extended

to flow-matching frameworks or autoregressive generative models.

A Roadmap to AP Bench

Underlying Rationale

Our rationale stems from a key observation: due to user proficiency level, the informativeness of prompts exerts a substantial impact on the quality of text-to-image (T2I) generation, while the community has paid limited attention to systematically addressing this issue. Motivated by this gap, our objective is to design a comprehensive and principled methodology and benchmark for evaluating the quality of prompts in T2I generation.

Inspired by recent preliminary work (Wang et al. 2025; Kim et al. 2022), we demonstrate how we quantify the informativeness of a text-to-image prompt P in conveying user intent Y to a generative model ϕ producing image X , using three carefully designed measures from an information-theoretic framework.

Mutual Information for User Intent Alignment We model T2I generation as a Markov chain $Y \rightarrow P \rightarrow X$ and characterize end-to-end prompt stability through the mutual information between user intent and generated images:

$$I(Y; X) = H(Y) - H(Y | X). \quad (1)$$

A larger $I(Y; X)$ indicates that more information about the user intent remains recoverable from the generated image. Because the true posterior $p(Y | X)$ is intractable, we use two complementary operational measures to evaluate intent preservation at different semantic granularities and the gap between user original prompt and optimized prompt.

CLIP Target-Category Probability (Mean and Standard Deviation). Following prior work (Du et al. 2023; Feng et al. 2022; Chefer et al. 2023), we approximate user intent via **entity–template expansion**. For each prompt P , we generate n images $\{I_1, \dots, I_n\}$ and classify them into a predefined category set $\mathcal{C}$ using CLIP. Let $a_j = q_{\text{CLIP}}(Y_c | I_j) \in [0, 1]$ denote the normalized CLIP probability assigned to the target user-intent category Y_c for image I_j . We then compute

$$\text{Acc}_{\text{mean}}(P) = \frac{1}{n} \sum_{j=1}^n a_j,$$
$$\text{Acc}_{\text{std}}(P) = \sqrt{\frac{1}{n} \sum_{j=1}^n (a_j - \text{Acc}_{\text{mean}}(P))^2}.$$

Here, $\text{Acc}_{\text{mean}}(P)$ measures the expected category-level recoverability of the user intent from generated images, while $\text{Acc}_{\text{std}}(P)$ measures how much this recoverability varies across stochastic generations from the same prompt. From an information-theoretic perspective, a higher target-category probability corresponds to lower surrogate uncertainty about Y_c given X , while a larger standard deviation indicates less stable intent transmission across samples.

MLLM Alignment Score. For finer-grained semantic evaluation, we use a multimodal LLM (MLLM) to score each generated image X_j against a short textual description

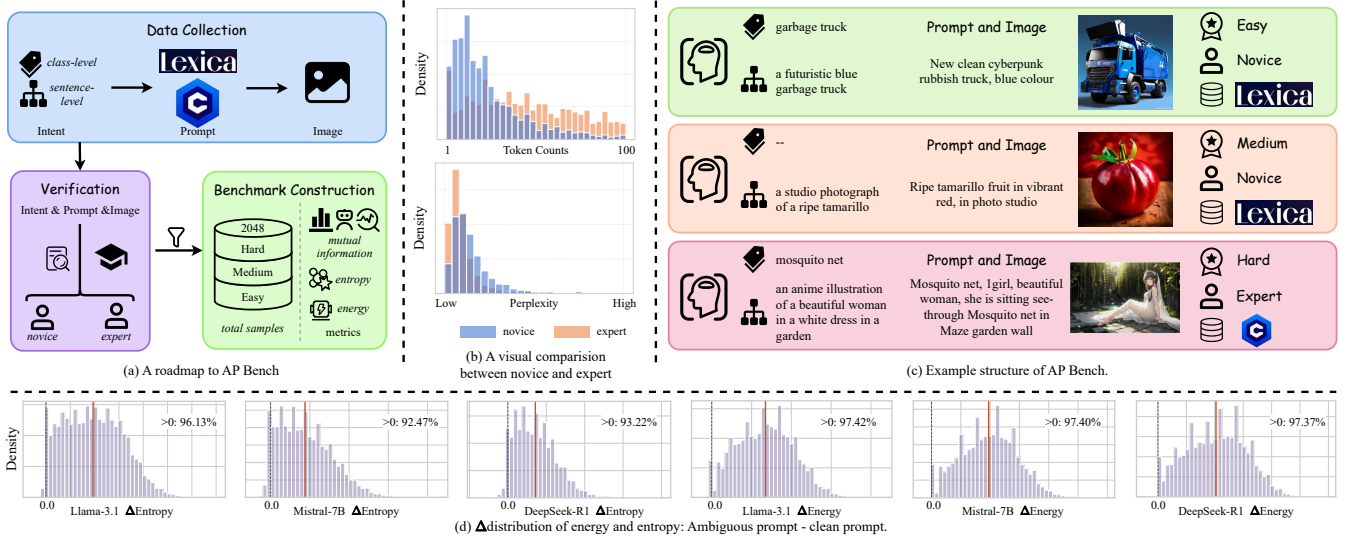


Figure 3: Overview and analysis of AP Bench. We construct AP Bench from 17,580 authentic prompt–image pairs and categorize samples by prompt proficiency and difficulty. Novice prompts exhibit higher uncertainty than expert prompts, while entropy and energy analysis on 42,000 matched prompt pairs further validates their effectiveness as prompt-side uncertainty indicators.

of Y , producing a soft score $s_j \in [0, 1]$. The average score is

$$S(P) = \frac{1}{n} \sum_{j=1}^n s_j.$$

This provides a continuous approximation of semantic alignment between user intent and generated images. More details on the connection between these metrics and mutual information are provided in Appendix A.

Prompt Entropy for T2I Informativeness Assessment By the data-processing inequality $I(Y; X) \leq I(Y; P)$, the maximum achievable stability is bounded by the information contained in the prompt, highlighting the importance of prompt optimization.

To quantify prompt informativeness (Farquhar et al. 2024; Cheng et al. 2025; Duan et al. 2023), we introduce **prompt entropy**. Intuitively, novice users often provide under-specified or ambiguous prompts, which are harder to interpret and may lead to unstable generation. In contrast, expert prompts are usually more specific and constrained, reducing uncertainty.

Formally, we define the T2I prompt entropy of P as

$$H(P) \approx -\frac{1}{T} \sum_{t=1}^T \log p_{\theta}(w_t \mid w_{<t}),$$

where p_{θ} is a pretrained language model. Lower entropy indicates that a prompt is more predictable and informative, and thus more likely to yield stable generations. See Appendix A for the derivation and its theoretical connection to $I(Y; X)$.

Prompt Energy for T2I Stability Assessment Existing T2I stability metrics, such as mutual information and prompt entropy, characterize end-to-end information transfer and prompt *aleatoric uncertainty*, but do not capture the model’s *epistemic uncertainty* (Ma et al. 2025). To complement them, we introduce **prompt energy**, which measures how confidently the model represents a prompt. For a prompt sequence

$x = (x_1, \dots, x_T)$, we define

$$E(x) = -\frac{1}{T} \sum_{t=1}^T z_t(x_t),$$

where $z_t(x_t)$ is the logit assigned to token x_t . Lower energy indicates higher model confidence and typically more stable generation, whereas higher energy suggests that the prompt contains less familiar concepts.

We further validate entropy and energy on over 42,000 matched ambiguous–clean prompt pairs from ATM (Du et al. 2023). As shown in Figure 3, clean prompts exhibit lower entropy and energy in more than 92% of cases, suggesting that these metrics provide complementary prompt-side signals for characterizing ambiguity and potential risks in intent transmission. They are used as *auxiliary diagnostics rather than absolute measures* of prompt quality.

Combining prompt-level entropy and energy with end-to-end metrics such as $I(Y; X)$ provides a more comprehensive characterization of generation stability, directly linking user-provided information to image fidelity. Implementation details, derivations from model logits, and the connection to classical Boltzmann energy are provided in Appendix A.

Benchmark Construction

Building on our information-theoretic formulation, we design a benchmark to empirically evaluate prompt informativeness and generation stability. Inspired by ImageNet (Russakovsky et al. 2015), we use its 1,000 semantic categories as the initial intent taxonomy and collect 17,580 authentic intent–prompt–image triplets from Lexica (<https://lexica.art/>) and Civitai (<https://civitai.com/>). Each triplet consists of a real user’s target intent, the prompt used to express that intent, and an image considered satisfactory by the user, together with the available generation parameters and auxiliary metadata.

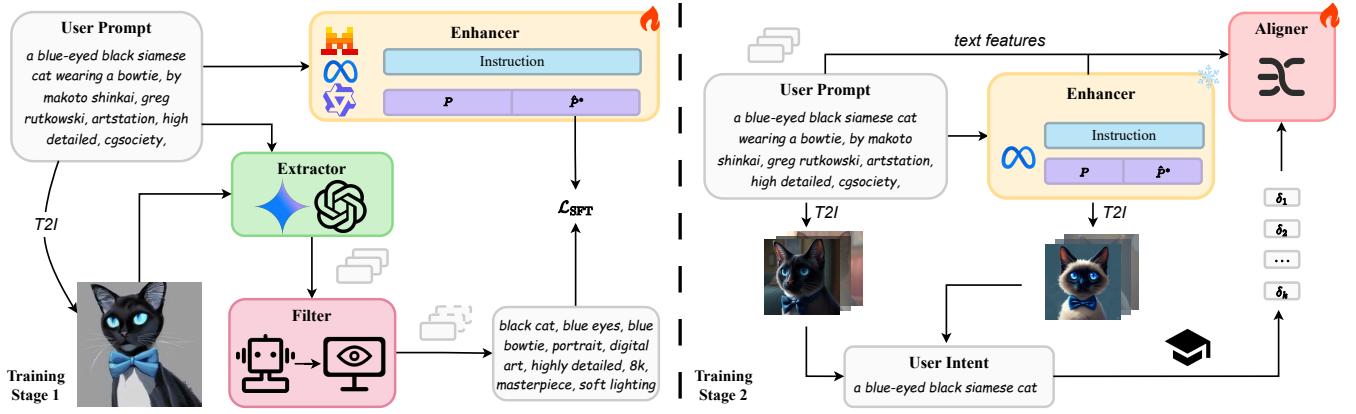


Figure 4: Overview of NoxEye training. It involves two stages, training enhancer and training aligner.

Following **manual review and re-verification**, the collected samples are carefully stratified into *novice* and *expert* subsets according to users’ proficiency in prompt design (Stanford University Information Technology 2026; Meta 2026; LetsEnhance 2026). This categorization considers the specificity, completeness, structural complexity, and use of generation-related syntax in each prompt. During this process, we further verify the correspondence among intent, prompt, and image, and filter the collected content to ensure ethical compliance, safety, and the absence of sensitive material. More Details are in Appendix A.

From the full collection, we select 2,048 representative samples for benchmark evaluation. In addition to intents drawn from the ImageNet categories, the benchmark includes 256 samples whose intents fall outside this predefined taxonomy, thereby extending the evaluation beyond closed-set category recognition. Based on the difficulty of generation stability, the benchmark is stratified into three levels: (1) **Easy**, where the user intent is explicit and the category belongs to one of the 1,000 predefined categories; (2) **Medium**, where the user intent remains explicit but the category falls outside the predefined set; and (3) **Hard**, where the category belongs to the predefined set but the user intent is expressed ambiguously.

This benchmark enables systematic evaluation of how prompt informativeness—quantified via mutual information, prompt entropy, and prompt energy—affects generation stability. Mutual information measures the **end-to-end intent preservation** between user intent and generated images, while prompt entropy and prompt energy serve as **auxiliary prompt-side measures** characterizing linguistic uncertainty and model compatibility, respectively. Together, these metrics provide a complementary evaluation framework: MI-based measures assess whether generated images preserve user intent, whereas entropy and energy capture linguistic ambiguity and representation uncertainty at the prompt level. To reduce evaluator bias, mutual information is estimated using CLIP (Radford et al. 2021) and Qwen3-VL (Bai, Cai et al. 2025), while prompt entropy and energy are measured with Llama-3.1-8B (Meta AI 2024), Mistral-7B-Instruct-v0.2 (Jiang et al. 2023), and DeepSeek-R1-Distill-Qwen-1.5B (DeepSeek-AI 2025). These complementary measurements provide fine-grained insights into how prompt characteristics influence

generation stability across different prompt proficiency levels.

NoxEye: An End-to-end Prompt Optimization Framework

NoxEye improves generation stability by selectively mapping user prompts into a model-aligned and better-organized prompt subspace. It consists of four components that sequentially construct reliable supervision, learn prompt refinement, and determine whether refinement is beneficial.

Information canonicalizer g_{can} . Given a user prompt P and its generated image X , the information canonicalizer constructs a semantically explicit and well-organized prompt for generation: $\hat{P}^* = g_{\text{can}}(P, X)$. The resulting prompt preserves the original intent while expressing visual information in a form that is more compatible with the target generative model.

Information filter g_{fil} . Since reconstructed prompts may contain visual hallucinations or semantic deviations, the information filter retains only candidates that preserve the user intent and provide valid information reorganization: $\mathcal{D} = \{(P, \hat{P}^*) \mid g_{\text{fil}}(P, \hat{P}^*, X, Y) = 1\}$, where Y denotes the user intent. The filtered pairs form the supervision data for prompt refinement.

Information enhancer π_θ . The information enhancer learns to transform the original prompt into the canonical prompt subspace: $P^* \sim \pi_\theta(\cdot \mid P)$. It is trained on $\mathcal{D}$ through supervised fine-tuning:

$$\mathcal{L}_{\text{SFT}}(\theta) = -\mathbb{E}_{(P, \hat{P}^*) \sim \mathcal{D}} \left[\log \pi_\theta(\hat{P}^* \mid P) \right].$$

Information aligner a_ψ . Prompt refinement is applied only when it is expected to provide a sufficient advantage under the target generative model ϕ . Given P , its refinement P^* , the model identity ϕ_m , and user intent Y , the aligner predicts

$$a_\psi(P, P^*, \phi_m, Y) = p_\psi(z = 1 \mid P, P^*, \phi_m, Y),$$

where $z = 1$ indicates that rewriting is beneficial. The supervision label is constructed by comparing the corresponding generations:

$$z = \mathbb{I}[Q(\phi_m(P^*), Y) - Q(\phi_m(P), Y) > \delta],$$

where Q measures generation quality with respect to the user intent and δ denotes the minimum required advantage.

$$P^{\text{out}} = \begin{cases} P^*, & a_\psi(P, P^*, \phi, Y) > \tau, \\ P, & \text{otherwise.} \end{cases}$$

where τ is a decision threshold that controls the minimum confidence required to apply prompt rewriting. Thus, NoxEye reorganizes prompt information while avoiding unnecessary or harmful rewriting.

Experiments

Settings

Data collection. NoxEye is trained in two stages. For training the information enhancer, we first select high-quality prompt–image pairs from our collected data, excluding all samples from *AP Bench*. For each user prompt, Gemini 2.5 (Comanici, Bieber et al. 2025) and GPT-4o (OpenAI et al. 2024) generate multiple candidate refinements. Each candidate is subsequently evaluated by human experts along three criteria: (i) semantic faithfulness to the original user prompt and alignment with the intended user intent, (ii) absence of hallucinated or unsupported content, and (iii) improvement in generation stability, measured by the consistency of images produced from the refined prompt. This process produces more than 11,000 original–refined prompt pairs for supervised fine-tuning.

For the information aligner, we construct over 4,000 samples containing the user intent, original prompt, and enhancer-refined prompt. Paired images are generated using *SD 2.1* (Rombach et al. 2022), *CoMat 2.1* (Jiang et al. 2024), *Stable Diffusion 3 (SD3)* (Esser et al. 2024), *Flux.1* (Labs 2024), *PixArt-Σ* (Chen, Ge et al. 2024), *Infinity (Inf)* (Han et al. 2025), and *Show-O2* (Xie, Yang, and Shou 2025). Qwen3-VL (Bai, Cai et al. 2025) evaluates image–intent consistency, and the score improvement provides the rewrite label. The aligner combines Qwen3 embeddings (Team 2025), target-model text embeddings, and lightweight linguistic features. The dataset is split into training and validation sets at an 8:2 ratio.

Implementation details. We train the information enhancer with *Mistral-7B* (Jiang et al. 2023), *Llama-3.1-8B* (Meta AI 2024), and *Qwen3-8B* (Team 2025), and use *Llama-3.1-8B* as the default backbone. Detailed configurations are provided in Appendix B. All experiments use AMD MI250X GPUs with fixed random seeds.

Comparative methods. We compare NoxEye with dedicated prompt optimization methods, including *Promptist* (Hao et al. 2023), *PAG* (Yun et al. 2025), and *PAE* (Mo et al. 2024); closed LLM rewriters, including *Gemini 2.0 Flash* (Pichai et al. 2024) and *DeepSeek-V4-Flash* (DeepSeek-AI et al. 2026); and open-weight LLM rewriters, including *Mistral-7B*, *Llama-3.1-8B*, and *Qwen3-8B*. All methods are evaluated under identical generation settings. Public implementations are used when available; *CoMat 2.1* is reproduced under the same experimental protocol.

Results

State-of-the-art Comparison on AP Bench. As shown in Tables 1 and 2, our method consistently surpasses prompt-

Method	SD2.1	CoMat	SD3	Flux.1	PixArt-Σ	Inf	Show-O2
Original Prompt	0.6544	0.6591	0.7379	0.7579	0.7536	0.7182	0.6847
Promptist (Hao et al. 2023)	0.5813 (-0.0731)	0.5876 (-0.0715)	0.6412 (-0.0967)	0.6569 (-0.1010)	0.6620 (-0.0916)	0.6335 (-0.0847)	0.6062 (-0.0785)
PAG (Yun et al. 2025)	0.4174 (-0.2370)	0.4245 (-0.2346)	0.4580 (-0.2799)	0.4754 (-0.2825)	0.4817 (-0.2719)	0.4550 (-0.2632)	0.4369 (-0.2478)
PAE (Mo et al. 2024)	0.6246 (-0.0298)	—	—	—	—	—	—
Mistral-7B (Jiang et al. 2023)	0.6435 (-0.0109)	0.6473 (-0.0118)	0.7224 (-0.0155)	0.7335 (-0.0244)	0.7434 (-0.0102)	0.7237 (+0.0055)	0.6963 (+0.0116)
Llama-3.1-8B (Meta AI 2024)	0.5736 (-0.0808)	0.5803 (-0.0788)	0.6441 (-0.0938)	0.6624 (-0.0955)	0.6713 (-0.0823)	0.6499 (-0.0683)	0.6175 (-0.0672)
Qwen3-8B (Team 2025)	0.6404 (-0.0140)	0.6442 (-0.0149)	0.7204 (-0.0175)	0.7418 (-0.0161)	0.7366 (-0.0170)	0.7149 (-0.0033)	0.6884 (+0.0037)
Gemini 2.0 Flash (Pichai et al. 2024)	0.6574 (+0.0030)	0.6633 (+0.0042)	0.7201 (-0.0178)	0.7461 (-0.0118)	0.7464 (-0.0072)	0.7156 (-0.0026)	0.6801 (-0.0046)
DeepSeek-V4-Flash (DeepSeek-AI et al. 2026)	0.6486 (-0.0058)	0.6534 (-0.0057)	0.7207 (-0.0172)	0.7416 (-0.0163)	0.7423 (-0.0113)	0.7133 (-0.0049)	0.6860 (+0.0013)
NoxEye (Ours) w/o Aligner	0.6493 (-0.0051)	0.6569 (-0.0022)	0.7266 (-0.0113)	0.7440 (-0.0139)	0.7512 (-0.0024)	0.7257 (+0.0075)	0.6963 (+0.0116)
NoxEye (Ours)	0.6613 (+0.0069)	0.6669 (+0.0078)	0.7412 (+0.0033)	0.7626 (+0.0047)	0.7594 (+0.0058)	0.7335 (+0.0153)	0.7014 (+0.0167)

Table 1: Evaluation results about alignment scores $\uparrow$ on AP Bench. Each cell reports the absolute score, with the **positive** and **negative** changes relative to the original prompt shown in parentheses. Bold values indicate the **best** results. AP Bench reveals whether prompt optimization methods genuinely improve intent preservation or instead introduce information loss under authentic user prompts.

Method	Energy $\downarrow$			Entropy $\downarrow$		
	Llama3	Mistral	DeepSeek	Llama3	Mistral	DeepSeek
Original Prompt	-10.2499	-8.3105	-9.6922	4.8349	3.2092	3.6316
Promptist (Hao et al. 2023)	-9.9156	-8.4298	-9.5479	5.0132	2.8596	3.8147
PAG (Yun et al. 2025)	-9.4217	-8.2512	-9.1560	5.3178	3.1739	3.9617
PAE (Mo et al. 2024)	-10.9813	-9.5547	-10.2767	4.6203	2.7240	3.6035
NoxEye (Ours)	-13.6701	-11.4979	-11.9008	3.7430	2.3791	3.2509

Table 2: Evaluation results for energy $\downarrow$ and entropy $\downarrow$ across different language models. Bold values indicate the **best** results.

optimization baselines across all evaluation metrics for all generative models. For challenges of all difficulty levels, our method achieves the best in mutual information, prompt energy and entropy. Notably, the results also reveal that prompt optimization does not universally improve upon the original user prompts. For many real-world prompts, excessive refinement may introduce redundant or distracting information, leading to neutral or even negative gains. In contrast, our method consistently yields positive improvements over the original prompts, demonstrating its ability to selectively enhance useful information while preserving the underlying user intent. Additional results are provided in Appendix B.

Boosting Performances on Existing T2I Models. To assess the contributions of the *Enhancer*, *Filter* and *Aligner* to generation quality and stability, we conduct a series of ablation experiments. As can be seen in Table 3, our results indicate that different generative models exhibit distinct prompt-preference spaces, which existing closed-source LLM-based optimizers fail to differentiate. In contrast, our method achieves higher alignment scores on the vast majority of generative models across all difficulty levels.

Evaluating Advanced Generative Ability. Beyond AP Bench, we further evaluate our method on the widely adopted GenEval and T2I-CompBench. As shown in Tables 4 and 5,

Enhancer	Filter	Aligner	SD2.1	CoMat	SD3	Flux.1	PixArt- Σ	Inf	Show-O2
✗	✗	✗	0.6544	0.6591	0.7379	0.7579	0.7536	0.7182	0.6847
✓	✗	✗	0.6523	0.6631	0.7179	0.7400	0.7418	0.7122	0.6852
✓	✓	✗	0.6493	0.6569	0.7266	0.7440	0.7512	0.7257	0.6963
✓	✓	○	0.6528	0.6585	0.7402	0.7581	0.7540	0.7211	0.6900
✓	✓	✓	0.6613	0.6669	0.7412	0.7626	0.7594	0.7335	0.7014

Table 3: Ablation results for the information enhancer, filter, and aligner about alignment score on AP Bench. The symbols ✓ and ✗ indicate enabled and disabled components, respectively, while ○ denotes a baseline in which DeepSeek-V4 (DeepSeek-AI et al. 2026) determines whether rewriting is necessary.

Method	Overall	Single Object	Two Objects	Counting	Colors	Color Attribute	Position
Original Prompt	0.5895	0.9875	0.6991	0.5067	0.8009	0.3579	0.1850
Promptist (Hao et al. 2023)	0.5871	0.9688	0.6898	0.4946	0.8024	0.3643	0.2025
PAG (Yun et al. 2025)	0.5662	0.9710	0.6641	0.4527	0.8024	0.3357	0.1711
PAE (Mo et al. 2024)	0.3858	0.8562	0.3914	0.2312	0.6809	0.0650	0.0900
NoxEye (Ours) w/o Aligner	0.5999	0.9741	0.6829	0.5098	0.8043	0.3668	0.2614
NoxEye (Ours)	0.5998	0.9871	0.7085	0.5089	0.8047	0.3746	0.2146

Table 4: Evaluation results about average scores↑ for all T2I models on GenEval (Ghosh, Hajishirzi, and Schmidt 2023). Bold values indicate **best**.

our approach achieves consistent and competitive improvements across all prompt optimization methods, with NoxEye attaining the best overall scores. Despite using synthetic prompts, both benchmarks reveal a consistent trend: rewriting offers limited benefits for simple, unambiguous prompts and may introduce harmful information, whereas its gains are more pronounced for complex prompts with multiple compositional constraints. These results demonstrate that our method is not limited to our self-constructed benchmark, but also generalizes well to established, community-recognized evaluation protocols, validating its robustness and practical effectiveness.

Human Evaluation on Prompting Informativeness. We conducted a user study with 20 trained experts with domain-specific professional backgrounds to compare our method with existing approaches from a human-centered perspective. Our approach was preferred most often, achieving scores of 0.44 (images) and 0.412 (prompts), compared to 0.417/0.143 and 0.366/0.222 for the baselines, respectively, demonstrating our method’s superior alignment with human preferences in Appendix B.

Inference Time Comparison. We further compare the inference efficiency of different prompt optimization methods. While Promptist/PAG and our method incur only marginal additional latency compared to the base SD2.1 model, PromptCoT (Yao et al. 2024) significantly increases inference time due to its multi-step reasoning process. Notably, our approach achieves a favorable balance between efficiency and performance, adding only a small inference-time overhead. Quantitative results are summarized in Table 6.

Qualitative Analysis.

Figure 5 illustrates additional visual examples, where we can observe **how NoxEye enhances prompt generation stabil-**

Method	Overall	Attribute Binding	Non Spatial	2D Spatial	3D Spatial	Numeracy	3-in-1
Original Prompt	0.3976	0.5586	0.3022	0.1290	0.3664	0.5528	0.3586
Promptist (Hao et al. 2023)	0.3797	0.5327	0.2944	0.1182	0.3486	0.5331	0.3390
PAG (Yun et al. 2025)	0.3661	0.5085	0.2929	0.1125	0.3481	0.5204	0.3139
NoxEye (Ours) w/o Aligner	0.3973	0.5578	0.2994	0.1258	0.3606	0.5676	0.3608
NoxEye (Ours)	0.3991	0.5601	0.3011	0.1291	0.3671	0.5623	0.3594

Table 5: Evaluation results about average scores↑ for all T2I models on T2I-CompBench (Huang et al. 2023). Bold values indicate **best**.

Method	SD 2.1	+Promptist/PAG	+PromptCoT	+Ours
Inf. Time (s)	0.0327	0.0358	4.6114	0.0748

Table 6: Inference time comparison across different methods.

ity. Specifically, unlike other prompt optimization methods that largely retain the original prompt structure (e.g., Amateur...) and ignore the user intent (e.g., black cat), NoxEye reconstructs the prompt in a more explicit manner (e.g., black cat ... a 2023 Ford F-150), clarifying the core intent while systematically enhancing relevant visual specifications. Thus, our method effectively refines the main content and provides detailed descriptions of artistic style, lighting, and other visual attributes. More results are in Appendix C.

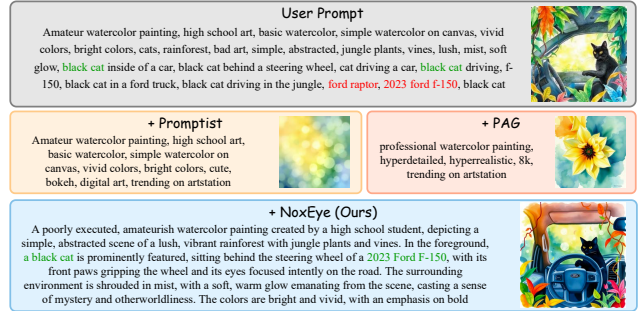


Figure 5: The generated images with the optimized prompts using our method. All images are generated by FLUX.1 (Labs 2024).

Conclusion

In this work, we study the problem of user prompting proficiency and its quantification and taxonomy issues in the T2I generation. To this end, we construct **AP Bench**, a benchmark consisting of novice and expert prompts to comprehensively evaluating the informativeness of the user prompts. Furthermore, we design **NoxEye**, a plug-and-play modular prompt optimization framework broadly compatible with diverse T2I models. Extensive experiments across AP Bench and additional benchmarks demonstrate that our approach substantially improves both image quality and stability, without introducing significant inference overhead. Overall, our contributions provide not only a practical method for prompt optimization, but also a novel perspective on modeling information flow in text-to-image generation. We believe this work lays a foundation for future research on principled evaluation and optimization of user-model interactions in generative systems.

References

- Bai, S.; Cai, Y.; et al. 2025. Qwen3-VL Technical Report. *arXiv:2511.21631*.
- Cao, T.; Wang, C.; Liu, B.; Wu, Z.; Zhu, J.; and Huang, J. 2023. Beautifulprompt: Towards automatic prompt engineering for text-to-image synthesis. *arXiv preprint arXiv:2311.06752*.
- Chefer, H.; Alaluf, Y.; Vinker, Y.; Wolf, L.; and Cohen-Or, D. 2023. Attend-and-excite: Attention-based semantic guidance for text-to-image diffusion models. *ACM transactions on Graphics (TOG)*, 42(4): 1–10.
- Chen, J.; Ge, C.; et al. 2024. PixArt- Σ : Weak – to – Strong Training of Diffusion Transformer for 4K Text – to – Image Generation. *arXiv* : 2403.04692.
- Chen, X.; Wu, Z.; Liu, X.; Pan, Z.; Liu, W.; Xie, Z.; Yu, X.; and Ruan, C. 2025. Janus-Pro: Unified Multimodal Understanding and Generation with Data and Model Scaling. *arXiv:2501.17811*.
- Cheng, D.; Huang, S.; Zhu, X.; Dai, B.; Zhao, W. X.; Zhang, Z.; and Wei, F. 2025. Reasoning with exploration: An entropy perspective. *arXiv preprint arXiv:2506.14758*.
- Comanici, G.; Bieber, E.; et al. 2025. Gemini 2.5: Pushing the Frontier with Advanced Reasoning, Multimodality, Long Context, and Next Generation Agentic Capabilities. *arXiv:2507.06261*.
- DeepSeek-AI. 2025. DeepSeek-R1: Incentivizing Reasoning Capability in LLMs via Reinforcement Learning. *arXiv:2501.12948*.
- DeepSeek-AI; et al. 2026. DeepSeek-V4: Towards Highly Efficient Million-Token Context Intelligence. *arXiv:2606.19348*.
- Du, C.; Li, Y.; Qiu, Z.; and Xu, C. 2023. Stable diffusion is unstable. *Advances in Neural Information Processing Systems (NeurIPS)*, 36: 58648–58669.
- Duan, J.; Cheng, H.; Wang, S.; Zavalny, A.; Wang, C.; Xu, R.; Kailkhura, B.; and Xu, K. 2023. Shifting attention to relevance: Towards the predictive uncertainty quantification of free-form large language models. *arXiv preprint arXiv:2307.01379*.
- Esser, P.; Kulal, S.; Blattmann, A.; Entezari, R.; Müller, J.; Saini, H.; Levi, Y.; Lorenz, D.; Sauer, A.; Boesel, F.; Podell, D.; Dockhorn, T.; English, Z.; Lacey, K.; Goodwin, A.; Marek, Y.; and Rombach, R. 2024. Scaling Rectified Flow Transformers for High-Resolution Image Synthesis. *arXiv:2403.03206*.
- Farquhar, S.; Kossen, J.; Kuhn, L.; and Gal, Y. 2024. Detecting hallucinations in large language models using semantic entropy. *Nature*, 630(8017): 625–630.
- Feng, W.; He, X.; Fu, T.-J.; Jampani, V.; Akula, A.; Narayana, P.; Basu, S.; Wang, X. E.; and Wang, W. Y. 2022. Training-free structured diffusion guidance for compositional text-to-image synthesis. *arXiv preprint arXiv:2212.05032*.
- Ghosh, D.; Hajishirzi, H.; and Schmidt, L. 2023. Geneval: An object-focused framework for evaluating text-to-image alignment. *Advances in Neural Information Processing Systems (NeurIPS)*, 36: 52132–52152.
- Goodfellow, I. J.; Pouget-Abadie, J.; Mirza, M.; Xu, B.; Warde-Farley, D.; Ozair, S.; Courville, A.; and Bengio, Y. 2014. Generative adversarial nets. *Advances in neural information processing systems (NeurIPS)*, 27.
- Han, J.; Liu, J.; Jiang, Y.; Yan, B.; Zhang, Y.; Yuan, Z.; Peng, B.; and Liu, X. 2025. Infinity: Scaling bitwise autoregressive modeling for high-resolution image synthesis. In *Proceedings of the Computer Vision and Pattern Recognition Conference (CVPR)*, 15733–15744.
- Hao, Y.; Chi, Z.; Dong, L.; and Wei, F. 2023. Optimizing prompts for text-to-image generation. *Advances in Neural Information Processing Systems (NeurIPS)*, 36: 66923–66939.
- Ho, J.; Jain, A.; and Abbeel, P. 2020. Denoising diffusion probabilistic models. *Advances in neural information processing systems (NeurIPS)*, 33: 6840–6851.
- Huang, K.; Duan, C.; Sun, K.; Xie, E.; Li, Z.; and Liu, X. 2025. T2I-CompBench++: An Enhanced and Comprehensive Benchmark for Compositional Text-to-Image Generation. *IEEE Transactions on Pattern Analysis Machine Intelligence (TPAMI)*, (01): 1–17.
- Huang, K.; Sun, K.; Xie, E.; Li, Z.; and Liu, X. 2023. T2i-compbench: A comprehensive benchmark for open-world compositional text-to-image generation. *Advances in Neural Information Processing Systems (NeurIPS)*, 36: 78723–78747.
- Jiang, A. Q.; Sablayrolles, A.; Mensch, A.; et al. 2023. Mistral 7B. *arXiv:2310.06825*.
- Jiang, D.; Song, G.; Wu, X.; Zhang, R.; Shen, D.; Zong, Z.; Liu, Y.; and Li, H. 2024. Comat: Aligning text-to-image diffusion model with image-to-text concept matching. *Advances in Neural Information Processing Systems (NeurIPS)*, 37: 76177–76209.
- Kalai, A. T.; Nachum, O.; Vempala, S. S.; and Zhang, E. 2025. Why Language Models Hallucinate. *arXiv:2509.04664*.
- Khan, M. A. H.; Jain, Y.; Bhattacharyya, S.; and Vineet, V. 2025. Test-time prompt refinement for text-to-image models. In *Proceedings of the IEEE/CVF International Conference on Computer Vision (ICCV)*, 6506–6516.
- Kim, J.-H.; Kim, Y.; Lee, J.; Yoo, K. M.; and Lee, S.-W. 2022. Mutual information divergence: A unified metric for multimodal generative models. *Advances in Neural Information Processing Systems*, 35: 35072–35086.
- Kim, S.; Cha, Y.; Yoo, J.; and Hong, S. 2025. Reward-agnostic prompt optimization for text-to-image diffusion models. *arXiv preprint arXiv:2506.16853*.
- Kingma, D. P.; and Welling, M. 2013. Auto-encoding variational bayes. *arXiv preprint arXiv:1312.6114*.
- Labs, B. F. 2024. FLUX. <https://github.com/black-forest-labs/flux>.
- Lee, D.; Kim, C.; Kim, S.; Cho, M.; and Han, W.-S. 2022. Autoregressive image generation using residual quantization. In *Proceedings of the IEEE/CVF conference on computer vision and pattern recognition (CVPR)*, 11523–11532.
- Lee, T.; Yasunaga, M.; et al. 2023. Holistic Evaluation of Text-to-Image Models. In Oh, A.; Naumann, T.; Globerson,

- A.; Saenko, K.; Hardt, M.; and Levine, S., eds., *Advances in Neural Information Processing Systems (NeurIPS)*, volume 36, 69981–70011. Curran Associates, Inc.
- LetsEnhance. 2026. How to Write Better AI Image Prompts: From Basic to Advanced [2026]. <https://letsenhance.io/blog/article/ai-text-prompt-guide/>. Accessed: 2026-07-22.
- Liu, M.; Ma, Y.; Yang, Z.; Dan, J.; Yu, Y.; Zhao, Z.; Hu, Z.; Liu, B.; and Fan, C. 2025. Llm4gen: Leveraging semantic representation of llms for text-to-image generation. In *Proceedings of the AAAI Conference on Artificial Intelligence (AAAI)*, volume 39, 5523–5531.
- Ma, H.; Pan, J.; Liu, J.; Chen, Y.; Zhou, J. T.; Wang, G.; Hu, Q.; Wu, H.; Zhang, C.; and Wang, H. 2025. Semantic Energy: Detecting LLM Hallucination Beyond Entropy. *arXiv preprint arXiv:2508.14496*.
- Mañas, O.; Astolfi, P.; Hall, M.; Ross, C.; Urbanek, J.; Williams, A.; Agrawal, A.; Romero-Soriano, A.; and Drozdal, M. 2024. Improving Text-to-Image Consistency via Automatic Prompt Optimization. *Trans. Mach. Learn. Res. (TMLR)*, 2024.
- Meta. 2026. Prompts for AI Images: 10 Examples and Tips for Better Results. <https://ai.meta.com/learn/ai-creativity/prompts-for-ai-images-10-examples-and-tips-for-better-results/>. Accessed: 2026-07-22.
- Meta AI. 2024. Introducing Llama 3.1: Our Most Capable Models to Date. <https://ai.meta.com/blog/meta-llama-3-1/>. Accessed: 2024-07-23.
- Mo, W.; Zhang, T.; Bai, Y.; Su, B.; Wen, J.-R.; and Yang, Q. 2024. Dynamic Prompt Optimizing for Text-to-Image Generation. In *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition*, 26627–26636.
- OpenAI; ; Hurst, A.; Lerer, A.; Goucher, A. P.; et al. 2024. GPT-4o System Card. *arXiv:2410.21276*.
- Pichai, S.; et al. 2024. Introducing Gemini 2.0: Our New AI Model for the Agentic Era. <https://blog.google/innovation-and-ai/models-and-research/google-deepmind/google-gemini-ai-update-december-2024/>. Google Blog, published December 11, 2024, accessed July 23, 2026.
- Radford, A.; Kim, J. W.; Hallacy, C.; Ramesh, A.; Goh, G.; Agarwal, S.; Sastry, G.; Askell, A.; Mishkin, P.; Clark, J.; et al. 2021. Learning transferable visual models from natural language supervision. In *International conference on machine learning (ICML)*, 8748–8763. PMLR.
- Radford, A.; Wu, J.; Child, R.; Luan, D.; Amodei, D.; Sutskever, I.; et al. 2019. Language models are unsupervised multitask learners. *OpenAI blog*, 1(8): 9.
- Ramesh, A.; Dhariwal, P.; Nichol, A.; Chu, C.; and Chen, M. 2022. Hierarchical text-conditional image generation with clip latents. *arXiv preprint arXiv:2204.06125*, 1(2): 3.
- Ramesh, A.; Pavlov, M.; Goh, G.; Gray, S.; Voss, C.; Radford, A.; Chen, M.; and Sutskever, I. 2021. Zero-shot text-to-image generation. In *International conference on machine learning (ICML)*, 8821–8831. PMLR.
- Rombach, R.; Blattmann, A.; Lorenz, D.; Esser, P.; and Ommer, B. 2022. High-Resolution Image Synthesis With Latent Diffusion Models. In *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR)*, 10684–10695.
- Rosenman, S.; Lal, V.; and Howard, P. 2023. Neuroprompts: An adaptive framework to optimize prompts for text-to-image generation. *arXiv preprint arXiv:2311.12229*.
- Russakovsky, O.; Deng, J.; Su, H.; Krause, J.; Satheesh, S.; Ma, S.; Huang, Z.; Karpathy, A.; Khosla, A.; Bernstein, M.; et al. 2015. Imagenet large scale visual recognition challenge. *International journal of computer vision (IJCV)*, 115(3): 211–252.
- Stability AI. 2023. Stable Diffusion Public Release. <https://stability.ai/blog/stable-diffusion-public-release>. Accessed: 2023-05-17.
- Stanford University Information Technology. 2026. GenAI Prompt Guide. <https://uit.stanford.edu/ai/prompt-guide>. Accessed: 2026-07-22.
- Team, Q. 2025. Qwen3 Technical Report. *arXiv:2505.09388*.
- Venkatesh, K.; Dalva, Y.; Lourentzou, I.; and Yanardag, P. 2025. RAVEL: Rare Concept Generation and Editing via Graph-driven Relational Guidance. *arXiv:2412.09614*.
- Wang, C.; Franzese, G.; Finamore, A.; Gallo, M.; and Michiardi, P. 2025. Information Theoretic Text-to-Image Alignment. In *The Thirteenth International Conference on Learning Representations (ICLR)*.
- Wu, M.; Wang, L.; Zhao, P.; Yang, F.; Zhang, J.; Liu, J.; Zhan, Y.; Han, W.; Sun, H.; Ji, J.; Sun, X.; Lin, Q.; Deng, W.; Zhang, D.; Sun, F.; and Ji, R. 2026. RePrompt: Reasoning-Augmented Reprompting for Text-to-Image Generation via Reinforcement Learning. In *The Fourteenth International Conference on Learning Representations (ICLR)*.
- Xie, J.; Yang, Z.; and Shou, M. Z. 2025. Show-o2: Improved Native Unified Multimodal Models. In *The Thirty-ninth Annual Conference on Neural Information Processing Systems (NeurIPS)*.
- Yang, H.; Zhou, Y.; Han, W.; and Shen, J. 2025. Self-Rewarding Large Vision-Language Models for Optimizing Prompts in Text-to-Image Generation. *arXiv preprint arXiv:2505.16763*.
- Yao, J.; Liu, Y.; Dong, Z.; Guo, M.; Hu, H.; Keutzer, K.; Du, L.; Zhou, D.; and Zhang, S. 2024. Promptcot: Align prompt distribution via adapted chain-of-thought. In *Proceedings of the IEEE/CVF conference on computer vision and pattern recognition (CVPR)*, 7027–7037.
- Yun, T.; Zhang, D.; Park, J.; and Pan, L. 2025. Learning to sample effective and diverse prompts for text-to-image generation. In *Proceedings of the Computer Vision and Pattern Recognition Conference (CVPR)*, 23625–23635.